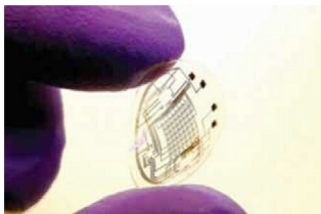


near-focus zone turns on and off in response to up and down movements of the user's head. A portable charger lets the user power emPower! for two-three days with a single charge.

## Bionic contact lenses

Researchers at Washington University have taken the first step towards creating digital contact lenses that can zoom in on distant objects and display useful facts. They have used manufacturing techniques at microscopic scales to combine a flexible, biologically safe contact lens with an imprinted electronic circuit and lights.



Digital contact lens with an imprinted electronic circuit and lights

The prototype device contains an electric circuit as well as red light-emitting diodes for a display, though it does not yet light up. Building the lenses was a challenge because materials that are safe for use in the body, such as the flexible organic materials used in contact lenses, are delicate. Manufacturing electrical circuits, however, involves inorganic materials, scorching temperatures and toxic chemicals.

Researchers built the circuits from layers of metal only a few nanometres thick—about one-thousandth the width of a human hair—and constructed light-emitting diodes one-third of a millimetre across. They then sprinkled the greyish powder of electrical components onto a sheet of flexible plastic. The shape of each tiny component dictates which piece it can attach to—a micro-fabrication technique known as self-assembly. Capillary forces—the forces that make water move up a plant's roots, and cause the edge of a glass of water to curve upward—pull the pieces into position.

## Classic antennae to harness more power from the sun

Professors Koby Scheuer, Yael Hanin and Amir Boag of Tel Aviv University's Department of Physical Electronics in Israel are developing a solar panel composed of nano-antennae instead of semiconductors.

Prof. Scheuer explained, "Both radio and optical waves are electromagnetic energy. When these waves are harvested, electrons are generated. These electrons can be converted into electric current. For optimal absorption, the antenna

## CAUTION NOTICE

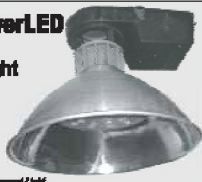
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All concerned in the Trade and Commerce and hereby requested to contact the undersigned address or any further details regarding the above and to avoid unnecessary litigation and/or legal proceedings.

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